

I. Theories of International Trade

(a) Haberler — Opportunity Cost Theory (Neo-Classical)

- Ricardo ke Labour Theory of Value ko hataya (1936).
- Cost of a good = jo doosra good chhodna pada (opportunity cost).
- Tool = PPC / Transformation Curve; slope = MRT.
- Alag-alag cost ratio → trade profitable. Same ratio → no trade.
- Equilibrium: Autarky mein MRT = MRS = domestic price ratio; trade ke baad = international price ratio.

PPC ka shape	Opportunity Cost	Specialization
Straight line	Constant	Complete
Concave to origin	Increasing	Incomplete (real world)
Convex to origin	Decreasing	Complete

(b) Heckscher-Ohlin Theory (Factor Endowment Theory)

- Heckscher (1919) + Ohlin (1933). Model = $2 \times 2 \times 2$.
- Dosh us good ko EXPORT karta hai jisme uska ABUNDANT factor intensively use ho.
- Labour-abundant desh (India) → labour-intensive goods export.
- Factor abundance: Price criterion (Ohlin) = P_K/P_L | Physical criterion (Leontief) = K/L .
- Assumptions: perfect competition, identical technology & tastes, CRS, no transport cost, no factor intensity reversal.

Theorem	Kya kehta hai
H-O Theorem	Abundant factor-intensive good export hoga
Factor Price Equalisation (Samuelson)	Trade se factor prices barabar; goods trade = factor mobility ka substitute
Stolper-Samuelson	Tariff se SCARCE factor ka real income badhta hai
Rybczynski	Ek factor ki supply ↑ → us factor-intensive good ka output ↑, doosre ka ↓

⚠ LEONTIEF PARADOX (1953) — Input-Output analysis se USA ka data test kiya. USA capital-abundant hote hue bhi labour-intensive goods export & capital-intensive import kar raha tha — H-O ka bilkul ULTA.
Reasons: US labour zyada skilled (human capital), natural resources ignore, demand reversal, tariff structure, factor intensity reversal.

Ricardo	Heckscher-Ohlin
Labour theory of value	Multi-factor (K & L)
Cost diff = technology/productivity	Cost diff = factor endowment
Partial equilibrium	General equilibrium
Technology different	Technology identical

2. Terms of Trade (TOT) & Gains from Trade

- TOT = export & import prices ka ratio. TOT improve → zyada gain.

Type of TOT	Formula	Proponent
Net Barter / Commodity (Tc)	$(P_x/P_m) \times 100$	Taussig
Gross Barter (Tg)	$(Q_m/Q_x) \times 100$	Taussig
Income TOT (Ty)	$(P_x/P_m) \times Q_x$	Dorrance
Single Factoral (Ts)	$(P_x/P_m) \times Z_x$	Viner
Double Factoral (Td)	$(P_x/P_m) \times (Z_x/Z_m)$	Viner

- Determination: **Reciprocal Demand** — J.S. Mill (TOT dono cost ratios ke beech); **Offer Curves** — Marshall & Edgeworth (intersection = equilibrium TOT).
- Factors: reciprocal demand, elasticity of demand, growth (export-biased se bigadti), tariff (large country → improve), devaluation (deteriorate), inflation.

PREBISCH-SINGER HYPOTHESIS — Developing countries ki TOT mein secular (long-run) DECLINE. Primary goods ki income elasticity kam, manufactured ki zyada. → Isse **Import Substitution** strategy ka tark aaya.

Gains from Trade

- Gain = domestic exchange ratio aur international ratio ka antar. Jitna zyada antar, utna zyada gain.
- 2 components: **Gain from Exchange** (consumption) + **Gain from Specialization** (production).
- Distribution: jiski demand kam elastic ho usko zyada gain; **chhota desh zyada gain** karta hai.
- Dynamic gains: economies of scale, technology transfer, capital inflow. Trade = **"Engine of Growth"** (Robertson); **Vent-for-Surplus** (Adam Smith / Myint).

3. Balance of Payments (BOP)

Balance of Trade	Balance of Payments
Sirf visible goods — narrow	Visible + invisible + capital — broad
Surplus/deficit ho sakta hai	Accounting mein hamesha BALANCE

- Structure: Current A/c (merchandise + invisibles + **unilateral transfers**) | Capital & Financial A/c (FDI, FPI, ECB, NRI deposits) | Reserves + **Errors & Omissions**.

Autonomous Items	Accommodating Items
Profit/business motive se	Gap bharne ke liye
"Above the line"	"Below the line"
Current + most capital a/c	Official reserve transactions

- BOP deficit ka matlab** = autonomous receipts < autonomous payments.
- Disequilibrium types: Cyclical, Structural, Secular, Temporary.
- Automatic adjustment: **Price-Specie Flow Mechanism** — David Hume (1752); flexible rate mein depreciation.
- Johnson: **Expenditure-SWITCHING** (devaluation, tariff) vs **Expenditure-REDUCING** (deflation, tight policy).
- Nurkse ki 3 shartein (equilibrium): no abnormal capital movement, no excessive trade restriction, full employment.

★ Teen Approaches to BOP Adjustment (MOST IMPORTANT)

	Elasticity	Absorption	Monetary
Proponent	Marshall, Lerner, Robinson	Sidney Alexander (1952)	Johnson, Mundell, Polak
Framework	Partial equilibrium	General (Keynesian)	Monetary, long-run
Focus	Relative prices	Income & absorption	Money demand & supply
Key tool	$Ex + Em > 1$	$B = Y - A$	$Ms = D + R$

MARSHALL-LERNER CONDITION: $Ex + Em > 1$ → tabhi devaluation se BOP sudhrega.

$Ex + Em = 1$ → no effect | $Ex + Em < 1$ → BOP aur bigadega.

J-CURVE EFFECT: Devaluation ke turant baad BOP pehle BIGADTA hai (short-run elasticities kam), phir SUDHARTA hai — graph "J" jaisa.

ABSORPTION APPROACH (Alexander): $B = Y - A$ ($A = C + I + G$)

Deficit → $A > Y$ (aamdani se zyada kharch). Surplus → $Y > A$.

Unemployment ho → devaluation se $Y \uparrow$ → kaam karega ✓ | Full employment ho → A kam karna padega ✗

MONETARY APPROACH (MABP): $Ms = D + R$ (D = domestic credit, R = reserves)

$Md > Ms$ → SURPLUS | $Ms > Md$ → DEFICIT. Deficit ka karan = excessive domestic credit creation.

Ilaaj = Domestic Credit Contraction (= IMF conditionality ka base). Disequilibrium self-correcting & temporary.

4. International Liquidity

- Components: Gold + Forex reserves + Reserve Tranche Position (IMF) + SDR.
- Problem: Inadequacy (trade tez badha, liquidity nahi) | Maldistribution (developed countries ke paas) | Confidence problem.

★ TRIFFIN DILEMMA (Robert Triffin, 1960)

Duniya ko liquidity chahiye → USA ko BOP deficit chalana padega → lekin lagatar deficit se dollar pe confidence girega.

Liquidity aur Confidence ek saath nahi mil sakte.

Result: Nixon Shock 1971 (gold convertibility khatam) → Bretton Woods collapse → floating rates 1973 se.

- SDR ("Paper Gold") — IMF ne 1969 mein banaya, pehli allocation 1970. Quota ke anupaat mein. Currency nahi — members ki currencies pe claim.
- Basket = 5 currencies: USD (~43%), Euro (~29%), Renminbi (2016 se added, ~12%), Yen, Pound. Review har 5 saal. Largest allocation = 2021, \$650 bn.

Plan	Proposal
Triffin Plan	IMF ko world central bank banao
Bernstein Plan	IMF quotas badhao
Stamp Plan	IMF certificates + developing countries ko aid ("link")
Keynes Plan (1944)	Int'l Clearing Union + currency "BANCOR"
White Plan	Jeeta → IMF & World Bank bane

5. Foreign Exchange Rate

- $NER = \text{market rate}$ | $RER = \text{inflation-adjusted}$ | $REER > 100 = \text{rupee OVERVALUED}$.
- Theories: **Mint Parity** (gold content ratio, obsolete) | **PPP** — **Gustav Cassel (1918)** | **BOP/Demand-Supply** (modern, most accepted) | **Interest Rate Parity** (covered/uncovered).
- Relative PPP: % change in exchange rate = **inflation rates ka antar**. Zyada inflation wali currency depreciate hogi. (Example: Big Mac Index)
- Systems: Fixed | Flexible | **Managed Float ("Dirty Float")** = **INDIA** | **Crawling peg** | **Currency board**.

★ IMPOSSIBLE TRINITY / TRILEMMA (Mundell-Fleming)

Ek desh ek saath ye teenon nahi rakh sakta — (1) Fixed exchange rate, (2) Free capital mobility, (3) Independent monetary policy. **Sirf DO chun sakte ho.**

Devaluation	Depreciation
Deliberate govt/RBI decision	Market forces se automatic
Fixed rate system mein	Flexible system mein
Opposite = Revaluation	Opposite = Appreciation

- India: Devaluations 1949, 1966, 1991. LERMS 1992 → unified rate 1993 → **Current a/c convertibility 1994** (IMF Art. VIII). Capital a/c partially convertible — **Tarapore Committee (1997, 2006)**.

6. Tariff & Non-Tariff Barriers

- By calculation: **Specific** (quantity pe) | **Ad valorem** (value ka %) | **Compound** | **Sliding scale**. Inflation mein specific duty ka protection ghat jata hai.
- Retaliatory: **CVD** (foreign subsidy offset), **Anti-dumping duty**, Retaliatory tariff.

EFFECTIVE RATE OF PROTECTION: $ERP = (t - a \cdot t_i) / (1 - a)$

Input pe tariff = 0 (ya $t_i < t$) → $ERP > \text{nominal rate}$. Agar $t_i > t$ → $ERP < t$ (negative bhi ho sakta hai).

Tariff Escalation: raw material pe kam, finished goods pe sabse zyada tariff → developing countries ke processed exports ko nuksan.

OPTIMUM TARIFF: $t^* = 1 / (e - 1)$ (e = foreign offer curve ki elasticity)

Chhota desh (price-taker) ka optimum tariff = ZERO. Bada desh TOT improve kar sakta hai, lekin retaliation se dono loser (Scitovsky).

⚠ **METZLER PARADOX** — Kabhi-kabhi tariff se import good ki **domestic price GIR jati hai** (badhne ki jagah), jab foreign demand bahut inelastic ho. → **Stolper-Samuelson ka ULTA result**.

Lerner Symmetry Theorem: Import tariff ka effect = Export tax ka effect.

- NTBs: Quota (tariff/unilateral/bilateral/mixing), Embargo, **VER**, Subsidies, **SPS & TBT**, Rules of Origin, local content requirement, govt procurement, exchange controls.

Tariff	Quota
PRICE mechanism	QUANTITY restriction
Revenue govt ko	Quota rent license-holder ko
Import quantity uncertain	Import quantity fixed
Corruption kam	Corruption zyada (license raj)
WTO mein ALLOWED	WTO mein PROHIBITED (Art. XI)

- **Tariffication** = Uruguay Round mein QRs ko tariffs mein convert karna (tariff zyada transparent hai).

7. Partial vs General Equilibrium (Tariff Analysis)

★ TARIFF KE 5 EFFECTS (Partial Equilibrium)

1. Protective/Production — domestic production $\uparrow$ | 2. Consumption — consumption $\downarrow$
3. Trade/Import — import $\downarrow$ | 4. Revenue — govt ko revenue | 5. Redistribution — CS se PS ko transfer

- Welfare: CS loss $(a+b+c+d) = PS$ gain $(a) + Govt$ revenue $(c) + Deadweight Loss $(b + d)$.
 $b =$ production DWL, $d =$ consumption DWL.$
- Small country ke liye tariff HAMESHA net welfare loss.
- Large country: TOT gain (e) milta hai $\rightarrow$ agar $e > b + d$ to net welfare GAIN (= optimum tariff ka logic).

Partial Equilibrium	General Equilibrium
Ek market, ceteris paribus	Poori economy, inter-related
Tool: Demand-Supply curves	Tool: PPC, CIC, Offer Curves
Juda $\rightarrow$ Marshall	Juda $\rightarrow$ Walras
Money/absolute prices	Relative prices

- Large country GE: tariff se apna offer curve andar shift $\rightarrow$ TOT improve. Retaliation se dono ke curve shift $\rightarrow$ trade volume ghat jata hai, dono ka nuksan.

8. Multilateral Bodies - WTO, IMF, World Bank

WTO

- ITO fail (Havana Charter 1948, US Congress ne ratify nahi kiya) $\rightarrow$ GATT (1947 sign, 1948 effective, 23 members) $\rightarrow$ WTO — 1 Jan 1995.
- HQ Geneva | ~166 members | DG Ngozi Okonjo-Iweala | Uruguay Round (1986–94) $\rightarrow$ Marrakesh Agreement 1994.
- GATT vs WTO: GATT = sirf agreement, sirf goods, weak dispute settlement. WTO = permanent institution, goods + services + IPR, binding DSB (negative consensus).
- Principles: MFN (Art. I), National Treatment (Art. III), transparency, S&DT.
- Agreements: GATT 1994 | GATS (4 modes; Mode 4 = India ka interest) | TRIPS (patent 20 yrs; India product patent 2005) | TRIMS | AoA | SPS, TBT | DSU.
- AoA ke 3 pillars: Market access, Domestic support, Export subsidies.
- Boxes: Green (allowed unlimited) | Blue (production-limiting, allowed) | Amber (trade-distorting, reduce; de minimis 10% developing, 5% developed).
- Doha Round (2001) deadlocked. Bali 2013 = TFA + Peace Clause. India ki demand = public stockholding ka permanent solution, SSM, 1986–88 reference price update. Appellate Body 2019 se paralysed.

9. IMF & World Bank

	IMF	World Bank
Focus	Monetary stability, exchange rate	Development, poverty reduction
Loan	Short-term, BOP crisis	Long-term, project/programme
Funds se	Member quotas	Market borrowing (bonds)
Head	Managing Director (European) – K. Georgieva	President (American) – Ajay Banga
Members / HQ	191 / Washington DC	IBRD 189 / Washington DC

- IMF quota = voting power (25% reserve tranche + 75% own currency). USA ~17.4% → de-facto VETO (85% majority rule). India ~2.75%.
- Surveillance = Article IV Consultation. Reports: WEO, GFSR, Fiscal Monitor. Criticism: conditionality / SAP / Washington Consensus.
- World Bank Group ke 5: IBRD (1944/45) | IDA (1960 – soft loan window, poorest) | IFC (1956 – private sector) | MIGA (1988 – political risk guarantee) | ICSID (1966 – dispute settlement).
- World Bank = IBRD + IDA only; World Bank Group = paanchon.

🔔 Exam Trap Alerts

- Concave PPC = INCOMPLETE specialization (complete nahi!)
- Stolper-Samuelson = SCARCE factor ka income badhta (abundant nahi!)
- Small country ka optimum tariff = ZERO
- BOP accounting mein hamesha balance, economic sense mein deficit
- WTO tariff ko pasand karta hai, quota ko nahi
- World Bank = IBRD + IDA only (Group = 5)

➤ Last-Minute Formula Box

Concept	Formula
Marshall-Lerner Condition	$Ex + Em > 1$
Absorption Approach	$B = Y - A \quad (A = C + I + G)$
Monetary Approach	$M_s = D + R$
Effective Rate of Protection	$ERP = (t - a_{ti}) / (1 - a)$
Optimum Tariff	$t^* = 1 / (e - 1)$
Net Barter TOT	$T_c = (P_x / P_m) \times 100$
Gross Barter TOT	$T_g = (Q_m / Q_x) \times 100$
Income TOT	$T_y = (P_x / P_m) \times Q_x$
Single Factoral TOT	$T_s = (P_x / P_m) \times Z_x$
Relative PPP	$\% \Delta ER = \text{Inflation}(\text{dom}) - \text{Inflation}(\text{for})$

★ "Kisne Kya Diya" - Quick Recall

Concept	Economist
Opportunity Cost Theory	Haberler
Factor Endowment Theory	Heckscher & Ohlin
Factor Price Equalisation	Samuelson
Input-Output Paradox (1953)	Leontief
Net Barter & Gross Barter TOT	Taussig
Income TOT	Dorrance
Factoral TOT	Viner
Reciprocal Demand	J.S. Mill
Offer Curves	Marshall & Edgeworth
Secular decline in TOT	Prebisch & Singer
Trade = "Engine of Growth"	D.H. Robertson
Vent-for-Surplus	Adam Smith / Myint
Price-Specie Flow Mechanism	David Hume
Absorption Approach	Sidney Alexander
Monetary Approach to BOP	Johnson & Mundell
Expenditure switching/reducing	H.G. Johnson
Liquidity vs Confidence Dilemma	Robert Triffin
"Bancor" / Clearing Union	J.M. Keynes
Purchasing Power Parity	Gustav Cassel
Impossible Trinity	Mundell & Fleming